Chapter 11 One-Pager — Dynamic Coalition Formation in Competitive FFA Games

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Chapters 2-10 all leaned on one crutch: a two-player game with an exact best-response and exploitability oracle, so "did it work?" was one number. Add a third player and alliances become possible — form, exploit, betray — while Nash turns both intractable and strategically empty, since it ignores coalitions entirely. Chapter 11 removes the crutch deliberately. This is the thesis frontier, not a consolidation step.

Approach. A native 4-player **So Long Sucker** engine — the 1950 game Nash, Shapley, Shubik and Hausner built to study alliances — implemented from the De Carufel & Jerade formalization, with a 2-player **minimax endgame** as its only exact anchor. On top: a **coalition detector** (help/harm from chip placement), **Shapley credit** redefined as the value of a member's win probability, a **coalition-aware MAPPO** trainer blending coalition and winner-takes-all reward by a weight `alpha`, and **EGTA + spinning-top** analysis reused from Chapter 10. **All numbers below are measured**, with one caveat: `scale_results.json` is a **pre-fix** run cited only as evidence of the bug below — the authoritative figures are `sweep_scale.json` (5 seeds) and post-fix `smoke_results.json`.

Key results (measured).

- *Certified wherever exactness exists.* 100/100 random games terminate, all rewards are zero-sum, `endgame_mismatches: 0` against exact minimax. The detector recovers a **planted `{0,1}` alliance** from the move stream alone (pair score **10.0**, cross-pair entries 0 or -1), and Shapley reproduces the glove and majority toys exactly, core emptiness included.
- *The step's biggest finding is a bug, and its lesson outlived it.* Two red checks shared one cause: ~**99.5% of random games end in deadlock**, so the winner falls to `_most_chips`, whose lowest-index tie-break gave seat 0 twice its fair share (all-random winners `[94,42,33,31]`). With an unbiased tie-break, symmetric Shapley spread fell **0.525 -> 0.013** and winners went uniform; the same fix deflated an impressive **0.87** hero win rate to an honest ~**0.41** against a 0.25 floor. In a game that nearly always ends near-tied, the tie-break rule is load-bearing.
- *"Coalitions don't emerge at scale" was overturned by finding the right knob.* `alpha` dominates and the **0.3 default sits in a dead zone**: at `alpha ~ 0` the proxy-credit gap is **+0.0376 +/- 0.0103** (~4.4x the sparse **0.0109**), while **every `alpha >= 0.3` cell is negative**. The effect *grows* with game size (~10x smoke), and the cheap critic-value proxy beats the expensive counterfactual credit (**+0.038 vs +0.013**).
- *Coalition behaviour is bought, not free.* `alpha = 0` drops hero win rate to ~**0.29**, essentially the random floor, while `alpha >= 0.1` holds ~**0.52**.
- *Strongly cyclic, honestly short of the bar.* The skill-ladder pool is transitive-dominant (cyclic **0.25-0.31**), but a coalition pool pushes cyclic to ~**0.57-0.69** — large, yet still failing the strict ">50% dominance" rule, so the check stays red. Harness: **4/5 PASS**.

Thesis connection. Contribution #1 generalises cleanly: opponent modelling lifts from "what kind of player is this" to "who is allied with whom", read straight off the moves. Contribution #2 gets its problem statement sharpened rather than solved — with an **empty core** and no Nash anchor, "safe" cannot mean bounded deviation from equilibrium and must become behavioural/population-based (piKL). Contribution #3 gains a working substitute for exploitability in EGTA plus the spinning-top decomposition, inheriting Chapter 10's caveat that the population you assemble decides whether you see a ladder or a wheel.

Open questions. Reconciling the engine's turn and tie-break model against De Carufel & Jerade is the highest-value follow-up — both the seat-0 bug and the red cyclic check trace to documented simplifications, and the theorems remain an unverified reading item. Whether a 3- or 4-player-aware EGTA projection surfaces the cycling the 2-type collapse discards. And what "safe" can mean at all when the core is empty.